

From Random Walks to Heat Kernels: An Introduction to Markov Chains

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Abstract

Markov chains provide a fundamental framework for describing stochastic systems whose future evolution depends on the present state but not on the past. This talk develops an elementary introduction to both discrete-time and continuous-time Markov chains, with particular emphasis on the connections between the two settings. We begin with transition matrices, the Chapman–Kolmogorov equations, simple random walks, stationary distributions, and convergence to equilibrium in finite state spaces. We then pass to continuous time by introducing exponential holding times and jump rates, leading naturally to the infinitesimal generator and the Kolmogorov equations. Finally, we interpret the transition kernel as a matrix exponential, connecting Markov-chain evolution with semigroup theory and the heat equation. The goal is to develop a unified picture of Markov chains in which transition matrices, generators, equilibrium distributions, and heat kernels arise as different descriptions of the same underlying stochastic evolution.

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1 Discrete-Time Markov Chains

1.1 The Markov property and transition matrices

Let S be a finite or countable set, called the *state space*. A stochastic process $(X_n)_{n \geq 0}$ with values in S is a *discrete-time Markov chain* if, whenever the conditional probabilities are defined,

$$\mathbb{P}(X_{n+1} = y \mid X_n = x, X_{n-1} = x_{n-1}, \dots, X_0 = x_0) = \mathbb{P}(X_{n+1} = y \mid X_n = x).$$

Thus, conditional on the present state, the future does not depend on the past. We assume throughout that the chain is *time-homogeneous*, meaning that the probability on the right does not depend on n . Its transition matrix is

$$P(x, y) = \mathbb{P}(X_{n+1} = y \mid X_n = x).$$

Every entry of P is nonnegative and every row sums to one. Such a matrix is called *stochastic*. We write $\mathbb{P}_x$ and $\mathbb{E}_x$ for probability and expectation conditional on $X_0 = x$.

Theorem 1.1 (Chapman–Kolmogorov). *For all integers $m, n \geq 0$ and states x, y ,*

$$P^{m+n}(x, y) = \sum_{z \in S} P^m(x, z) P^n(z, y).$$

Equivalently, the $(m + n)$ -step transition matrix is the matrix product $P^{m+n} = P^m P^n$.

Proof. Partition according to the value of the intermediate state X_m . The law of total probability gives

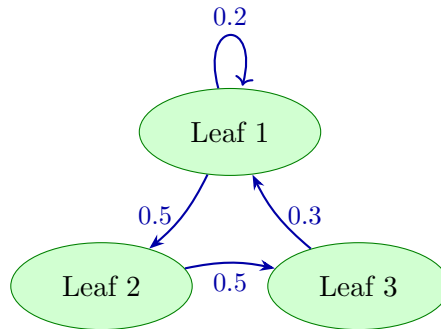
$$\begin{aligned} \mathbb{P}_x(X_{m+n} = y) &= \sum_{z \in S} \mathbb{P}_x(X_m = z, X_{m+n} = y) \\ &= \sum_{z \in S} \mathbb{P}_x(X_m = z) \mathbb{P}_x(X_{m+n} = y \mid X_m = z). \end{aligned}$$

By repeated application of the Markov property and time homogeneity, the last conditional probability is the probability of going from z to y in n steps, namely $P^n(z, y)$. Similarly, $\mathbb{P}_x(X_m = z) = P^m(x, z)$. Substitution yields the claimed sum. The sum is exactly the (x, y) entry of the matrix product $P^m P^n$. $\square$

Example 1.2 (A frog on three lotus leaves). Suppose a frog jumps once per minute among three leaves according to

$$P = \begin{pmatrix} 0.2 & 0.5 & 0.3 \\ 0.4 & 0.1 & 0.5 \\ 0.3 & 0.3 & 0.4 \end{pmatrix}.$$

If it is on leaf 1, it stays there with probability 0.2, jumps to leaf 2 with probability 0.5, and jumps to leaf 3 with probability 0.3. The entry $P^n(i, j)$ is the probability that the frog is on leaf j after n minutes when it starts on leaf i .



1.2 Simple random walk

Let $(\xi_i)_{i \geq 1}$ be independent random variables with

$$\mathbb{P}(\xi_i = 1) = p, \quad \mathbb{P}(\xi_i = -1) = q = 1 - p,$$

and set

$$X_n = X_0 + \sum_{i=1}^n \xi_i.$$

This is a Markov chain on $\mathbb{Z}$: from x it moves to $x+1$ with probability p and to $x-1$ with probability q .

Proposition 1.3 (Distribution of simple random walk). *If $X_0 = 0$, then for $|k| \leq n$ with $k \equiv n \pmod{2}$,*

$$\mathbb{P}_0(X_n = k) = \binom{n}{(n+k)/2} p^{(n+k)/2} q^{(n-k)/2}.$$

For all other integers k , the probability is zero.

Proof. Suppose that among the first n increments there are u upward steps and d downward steps. Then $u + d = n$ and $X_n = u - d$. The event $X_n = k$ therefore requires

$$u = \frac{n+k}{2}, \quad d = \frac{n-k}{2}.$$

These are nonnegative integers exactly when $|k| \leq n$ and $k \equiv n \pmod{2}$. There are $\binom{n}{u}$ possible locations for the upward steps. Independence gives probability $p^u q^d$ for each such sequence, proving the formula. If the parity or range condition fails, no sequence of n increments can have sum k . $\square$

Proposition 1.4 (First two moments). *For a walk starting from x ,*

$$\mathbb{E}_x[X_n] = x + n(p - q), \quad \text{Var}_x(X_n) = 4npq.$$

Proof. The increment satisfies

$$\mathbb{E}[\xi_i] = p - q, \quad \mathbb{E}[\xi_i^2] = 1,$$

and hence

$$\text{Var}(\xi_i) = 1 - (p - q)^2 = 1 - (p^2 - 2pq + q^2) = 4pq,$$

where we used $p + q = 1$. Linearity of expectation gives

$$\mathbb{E}_x[X_n] = x + \sum_{i=1}^n \mathbb{E}[\xi_i] = x + n(p - q).$$

Because the increments are independent, their variances add, so

$$\text{Var}_x(X_n) = \sum_{i=1}^n \text{Var}(\xi_i) = 4npq.$$

$\square$

1.3 Stationary distributions

From this point through the end of this section, assume that S is finite. A probability row vector π on S is *stationary* if

$$\pi P = \pi.$$

If X_0 has distribution π , then X_n has distribution $\pi P^n = \pi$ for every n .

A chain is *irreducible* if for every $x, y \in S$ there is some $n \geq 0$ such that $P^n(x, y) > 0$.

Theorem 1.5 (Existence and uniqueness of equilibrium). *Every finite irreducible Markov chain has a unique stationary distribution π , and $\pi(x) > 0$ for every $x \in S$.*

Proof. Fix a probability distribution μ and form the Cesàro averages

$$\nu_N = \frac{1}{N} \sum_{k=0}^{N-1} \mu P^k.$$

The probability simplex in $\mathbb{R}^{|S|}$ is compact, so some subsequence ν_{N_j} converges to a probability vector π . Moreover,

$$\nu_N P - \nu_N = \frac{\mu P^N - \mu}{N}.$$

Every coordinate of the numerator has absolute value at most 1, so the right-hand side converges to zero. Taking the limit along the subsequence and using continuity of matrix multiplication gives $\pi P = \pi$. Thus a stationary distribution exists.

At least one coordinate, say $\pi(z)$, is positive. For any y , irreducibility provides an n with $P^n(z, y) > 0$. Since $\pi P^n = \pi$,

$$\pi(y) = \sum_{x \in S} \pi(x) P^n(x, y) \geq \pi(z) P^n(z, y) > 0.$$

Thus every coordinate is positive.

It remains to prove uniqueness. Let π and ν be stationary distributions. By the preceding argument both are strictly positive. Set

$$c = \min_{x \in S} \frac{\pi(x)}{\nu(x)}, \quad \rho = \pi - c\nu.$$

Then $\rho \geq 0$, $\rho P = \rho$, and $\rho(x_0) = 0$ for at least one state x_0 attaining the minimum. For any y , irreducibility supplies an n with $P^n(y, x_0) > 0$. Since $\rho P^n = \rho$,

$$0 = \rho(x_0) = \sum_{z \in S} \rho(z) P^n(z, x_0) \geq \rho(y) P^n(y, x_0).$$

Both factors on the right are nonnegative and the second is positive, so $\rho(y) = 0$. Hence $\rho = 0$ and $\pi = c\nu$. Summing the coordinates gives $1 = c$, and therefore $\pi = \nu$. $\square$

1.4 Convergence to equilibrium

For a state x , define its period by

$$d(x) = \gcd\{n \geq 1 : P^n(x, x) > 0\}.$$

In an irreducible chain all states have the same period. The chain is *aperiodic* if this common period is 1.

Lemma 1.6 (Eventual positivity). *If a finite Markov chain is irreducible and aperiodic, then there is an integer N such that $P^n(x, y) > 0$ for every pair x, y and every $n \geq N$.*

Proof. Fix a state x . The positive return times

$$A_x = \{n \geq 1 : P^n(x, x) > 0\}$$

have greatest common divisor 1. Consequently, there are finitely many $a_1, \dots, a_\ell \in A_x$ whose greatest common divisor is 1: starting with any element of A_x and repeatedly taking the gcd with another element, the decreasing sequence of positive gcds must eventually reach the gcd of the whole set.

We now show that every sufficiently large integer is a nonnegative integer combination of $a_1, \dots, a_\ell$. Work modulo a_1 . The set of residues of all such nonnegative combinations is closed under addition. Because the ambient group $\mathbb{Z}/a_1\mathbb{Z}$ is finite, this set is a subgroup: if a residue r occurs, then some positive multiple kr is zero, so $(k-1)r = -r$ also occurs. The subgroup generated by the residues of $a_1, \dots, a_\ell$ is the whole group because their gcd is 1. For each residue r , choose a nonnegative combination b_r having residue r . If $M_x = \max_r b_r$ and $n \geq M_x$ has residue r , then $n = b_r + ka_1$ for some $k \geq 0$. Thus n is a nonnegative combination of the a_i .

For $a, b \in A_x$, Chapman–Kolmogorov gives $P^{a+b}(x, x) \geq P^a(x, x)P^b(x, x) > 0$. It follows that $P^n(x, x) > 0$ for every $n \geq M_x$.

Now fix y . Irreducibility gives a path from x to y of some length r_{xy} with $P^{r_{xy}}(x, y) > 0$. Whenever $n - r_{xy} \geq M_x$, Chapman–Kolmogorov yields

$$P^n(x, y) \geq P^{n-r_{xy}}(x, x)P^{r_{xy}}(x, y) > 0.$$

There are only finitely many ordered pairs (x, y) , so the maximum of the finitely many thresholds $M_x + r_{xy}$ is a common N with the desired property. $\square$

For probability distributions μ and ν on S , their total variation distance is

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_{x \in S} |\mu(x) - \nu(x)|.$$

Theorem 1.7 (Finite-state convergence theorem). *If a finite Markov chain is irreducible and aperiodic, with stationary distribution π , then*

$$\lim_{n \rightarrow \infty} P^n(x, y) = \pi(y) \quad \text{for every } x, y \in S.$$

In fact, the convergence is geometric in total variation.

Proof. By the preceding lemma, choose N such that the matrix $A = P^N$ has all entries positive. Let $s = |S|$,

$$\varepsilon = \min_{x, y} A(x, y) > 0, \quad \beta = s\varepsilon.$$

Because each row of A sums to one, $0 < \beta \leq 1$. Let u be the uniform distribution on S . If $\beta < 1$, define

$$R(x, y) = \frac{A(x, y) - \varepsilon}{1 - \beta}.$$

The matrix R is stochastic, and

$$A(x, \cdot) = \beta u + (1 - \beta)R(x, \cdot).$$

If $\beta = 1$, every row of A equals u and the following contraction holds with factor zero. Otherwise, for any distributions μ, ν ,

$$\mu A - \nu A = (1 - \beta)(\mu R - \nu R).$$

Every stochastic matrix is nonexpansive in total variation: indeed,

$$\begin{aligned} \|\mu R - \nu R\|_{\text{TV}} &= \frac{1}{2} \sum_y \left| \sum_x (\mu(x) - \nu(x)) R(x, y) \right| \\ &\leq \frac{1}{2} \sum_{x, y} |\mu(x) - \nu(x)| R(x, y) = \|\mu - \nu\|_{\text{TV}}. \end{aligned}$$

Consequently,

$$\|\mu A - \nu A\|_{\text{TV}} \leq (1 - \beta) \|\mu - \nu\|_{\text{TV}}.$$

Iterating gives

$$\|\mu A^k - \nu A^k\|_{\text{TV}} \leq (1 - \beta)^k \|\mu - \nu\|_{\text{TV}}.$$

Write $n = kN + r$ with $0 \leq r < N$. Since $\pi P = \pi$, we have $\pi A^k = \pi$. Taking $\mu = \delta_x P^r$ and $\nu = \pi$ gives

$$\|\delta_x P^n - \pi\|_{\text{TV}} = \|(\delta_x P^r) A^k - \pi A^k\|_{\text{TV}} \leq (1 - \beta)^k.$$

The right-hand side tends to zero as n , and hence k , tends to infinity. For every y , $|P^n(x, y) - \pi(y)| \leq 2\|\delta_x P^n - \pi\|_{\text{TV}}$, which proves the stated pointwise convergence as well as geometric convergence. $\square$

Remark 1.8. Aperiodicity is necessary for ordinary pointwise convergence. For example, simple random walk on a bipartite graph alternates between the two sides and therefore cannot settle to the stationary distribution at every time.

2 Continuous-Time Markov Chains

2.1 Poissonizing a discrete chain

Let $T_1, T_2, \dots$ be independent exponential random variables of rate $r > 0$, and let $(\Phi_k)_{k \geq 0}$ be a Markov chain with transition matrix P , independent of the T_i . Define

$$S_0 = 0, \quad S_k = \sum_{i=1}^k T_i, \quad N_t = \max\{k : S_k \leq t\},$$

and set $X_t = \Phi_{N_t}$. Equivalently, $X_t = \Phi_k$ when $S_k \leq t < S_{k+1}$. Conditional on $N_t = k$,

$$\mathbb{P}_x(X_t = y \mid N_t = k) = P^k(x, y).$$

Proposition 2.1 (The jump count is Poisson). *For every $t \geq 0$,*

$$\mathbb{P}(N_t = k) = e^{-rt} \frac{(rt)^k}{k!}, \quad k = 0, 1, 2, \dots$$

Thus N_t has the Poisson distribution with mean rt .

Proof. For $k = 0$,

$$\mathbb{P}(N_t = 0) = \mathbb{P}(T_1 > t) = e^{-rt}.$$

For $k \geq 1$, the sum S_k has the gamma density

$$f_{S_k}(s) = \frac{r^k s^{k-1} e^{-rs}}{(k-1)!}, \quad s > 0.$$

This follows by induction using convolution: it is the exponential density when $k = 1$, and convolving the displayed density with $re^{-r(t-s)}$ changes the power from $k-1$ to k and the normalizing factor from $(k-1)!$ to $k!$. Now $N_t = k$ exactly when $S_k \leq t < S_k + T_{k+1}$. Independence of S_k and T_{k+1} gives

$$\begin{aligned} \mathbb{P}(N_t = k) &= \int_0^t \mathbb{P}(T_{k+1} > t-s) f_{S_k}(s) ds \\ &= \int_0^t e^{-r(t-s)} \frac{r^k s^{k-1} e^{-rs}}{(k-1)!} ds \\ &= e^{-rt} \frac{r^k}{(k-1)!} \int_0^t s^{k-1} ds = e^{-rt} \frac{(rt)^k}{k!}. \end{aligned}$$

This includes every $k \geq 1$ and, together with the $k = 0$ calculation, proves the result. $\square$

2.2 Jump rates and the generator

Consider a frog in a pond with many lotus leaves, each equipped with an independent buzzer. Suppose the frog is initially at leaf x . For each leaf $y \neq x$, the buzzer at y rings at rate $q(x, y)$, and the frog jumps to the leaf whose buzzer rings first. The total jump rate from x is

$$q(x) = \sum_{y \neq x} q(x, y).$$

Proposition 2.2 (Competing exponentials). *The holding time at x is exponential with rate $q(x)$, and, when $q(x) > 0$,*

$$\mathbb{P}(\text{next state is } y \mid \text{current state is } x) = \frac{q(x, y)}{q(x)}.$$

Proof. Write $T_y \sim \text{Exp}(q(x, y))$ for the independent clocks and $T = \min_{y \neq x} T_y$. For $t \geq 0$,

$$\mathbb{P}(T > t) = \prod_{y \neq x} \mathbb{P}(T_y > t) = \prod_{y \neq x} e^{-q(x, y)t} = e^{-q(x)t}.$$

This is the survival function of an exponential variable of rate $q(x)$. For a fixed $y \neq x$, condition on $T_y = t$. Since exponential variables with positive rates have continuous distributions, ties have probability zero, and independence gives

$$\begin{aligned} \mathbb{P}(T_y = T) &= \int_0^\infty q(x, y) e^{-q(x, y)t} \prod_{z \neq x, y} e^{-q(x, z)t} dt \\ &= \int_0^\infty q(x, y) e^{-q(x)t} dt = \frac{q(x, y)}{q(x)}. \end{aligned}$$

If some rate is zero, its clock may be interpreted as infinite and the same formula remains valid. $\square$

The *generator* is the matrix

$$Q(x, y) = \begin{cases} q(x, y), & y \neq x, \\ -q(x), & y = x. \end{cases}$$

Its off-diagonal entries are nonnegative and each row sums to zero.

2.3 Uniformization

Theorem 2.3 (Uniformization). *Let S be finite and let Q be a generator. Choose $r \geq \max_x q(x)$ and define*

$$P(x, y) = \frac{q(x, y)}{r} \quad (y \neq x), \quad P(x, x) = 1 - \frac{q(x)}{r}.$$

Then P is stochastic and $Q = r(P - I)$. If a rate- r Poisson clock is used to apply successive P -steps, the resulting continuous-time chain has generator Q .

Proof. The entries of P are nonnegative because $r \geq q(x)$, and for each x ,

$$\sum_y P(x, y) = 1 - \frac{q(x)}{r} + \sum_{y \neq x} \frac{q(x, y)}{r} = 1.$$

Thus P is stochastic. For $y \neq x$, $r(P - I)(x, y) = rP(x, y) = q(x, y) = Q(x, y)$, while

$$r(P - I)(x, x) = r \left(1 - \frac{q(x)}{r} - 1 \right) = -q(x) = Q(x, x).$$

Hence $Q = r(P - I)$.

To identify the generator of the construction, let $h \downarrow 0$. A rate- r Poisson process has one event in $[0, h]$ with probability $rh + o(h)$ and at least two events with probability $o(h)$. Therefore, for $y \neq x$,

$$\mathbb{P}_x(X_h = y) = (rh + o(h))P(x, y) + o(h) = q(x, y)h + o(h).$$

Also,

$$\mathbb{P}_x(X_h = x) = 1 - q(x)h + o(h).$$

Thus

$$\lim_{h \downarrow 0} \frac{\mathbb{P}_x(X_h = y) - \mathbf{1}_{\{x=y\}}}{h} = Q(x, y),$$

which is precisely the definition of the infinitesimal generator. A diagonal P -step is a virtual jump: the Poisson clock rings but the state does not change. $\square$

Theorem 2.4 (Stationarity under uniformization). *If $Q = r(P - I)$, then a probability vector π satisfies $\pi P = \pi$ if and only if it satisfies $\pi Q = 0$.*

Proof. Direct calculation gives

$$\pi Q = r\pi(P - I) = r(\pi P - \pi).$$

Since $r > 0$, the right-hand side is zero exactly when $\pi P - \pi = 0$. $\square$

Theorem 2.5 (Eigenfunctions under uniformization). *Suppose $Q = r(P - I)$. A column vector φ satisfies $P\varphi = \lambda\varphi$ if and only if*

$$Q\varphi = r(\lambda - 1)\varphi.$$

Proof. If $P\varphi = \lambda\varphi$, then

$$Q\varphi = r(P - I)\varphi = r(\lambda\varphi - \varphi) = r(\lambda - 1)\varphi.$$

Conversely, if the displayed equation holds, divide by r and rearrange to obtain $P\varphi = \lambda\varphi$. $\square$

2.4 The heat kernel and Kolmogorov equations

Define the transition matrix, or *heat kernel*, at time t by

$$H_t(x, y) = \mathbb{P}_x(X_t = y).$$

Theorem 2.6 (Heat-kernel formula). *For a finite-state continuous-time Markov chain with generator Q ,*

$$H_t = e^{tQ} := \sum_{k=0}^{\infty} \frac{t^k Q^k}{k!}.$$

Proof. Choose a uniformization rate $r \geq \max_x q(x)$ and put $P = I + Q/r$. Conditional on the Poisson count N_t , the chain has made N_t steps of P . The law of total probability and the Poisson formula therefore give

$$\begin{aligned} H_t(x, y) &= \sum_{k=0}^{\infty} P^k(x, y) e^{-rt} \frac{(rt)^k}{k!} \\ &= \left(e^{-rt} \sum_{k=0}^{\infty} \frac{(rtP)^k}{k!} \right) (x, y) \\ &= (e^{-rtI} e^{rtP})(x, y). \end{aligned}$$

The matrices $-rtI$ and rtP commute, so their exponentials multiply to the exponential of their sum. Hence

$$H_t = e^{rt(P-I)} = e^{tQ}.$$

The matrix power series converges absolutely in any matrix norm, justifying the displayed rearrangements. $\square$

Theorem 2.7 (Kolmogorov equations and the semigroup law). *The heat kernel satisfies*

$$\frac{d}{dt} H_t = QH_t = H_t Q, \quad H_0 = I,$$

and, for $s, t \geq 0$,

$$H_{s+t} = H_s H_t.$$

Proof. On every bounded interval, the power series for e^{tQ} and its termwise derivative converge uniformly. Therefore

$$\frac{d}{dt} e^{tQ} = \sum_{k=1}^{\infty} \frac{k t^{k-1} Q^k}{k!} = Q \sum_{j=0}^{\infty} \frac{t^j Q^j}{j!} = Q e^{tQ}.$$

Since every power of Q commutes with Q , the same calculation also gives $e^{tQ} Q$. At $t = 0$ the series equals I .

Finally, sQ and tQ commute, and the product of their absolutely convergent power series may be regrouped by total degree:

$$\begin{aligned} e^{sQ} e^{tQ} &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n \frac{s^k t^{n-k}}{k!(n-k)!} \right) Q^n \\ &= \sum_{n=0}^{\infty} \frac{(s+t)^n Q^n}{n!} = e^{(s+t)Q}. \end{aligned}$$

Using $H_u = e^{uQ}$ proves the semigroup law. $\square$

The analogy with the classical heat equation is now explicit. The equation $\partial_t u = \Delta u$ has solution $u(t) = e^{t\Delta}u(0)$, while a Markov chain evolves by $H_t = e^{tQ}$; the generator Q plays the role of a discrete Laplacian.

2.5 Long-time convergence in continuous time

A finite-state generator Q is *irreducible* if for every x, y there is a sequence $x = x_0, x_1, \dots, x_m = y$ with $q(x_{i-1}, x_i) > 0$ for every i .

Theorem 2.8 (Continuous-time convergence theorem). *Let Q be an irreducible generator on a finite state space. There is a unique probability distribution π satisfying $\pi Q = 0$, it is strictly positive, and*

$$\lim_{t \rightarrow \infty} H_t(x, y) = \pi(y) \quad \text{for every } x, y \in S.$$

Proof. Choose $a > \max_x q(x)$ and define

$$\tilde{P} = I + \frac{1}{a}Q.$$

Uniformization shows that $\tilde{P}$ is stochastic. It is irreducible because every positive-rate path for Q is a positive-probability path for $\tilde{P}$. Moreover,

$$\tilde{P}(x, x) = 1 - \frac{q(x)}{a} > 0$$

for every x . A return in one step is therefore possible at every state, so the chain is aperiodic.

By the finite-state existence and uniqueness theorem, $\tilde{P}$ has a unique strictly positive stationary distribution π . The stationarity under uniformization theorem gives

$$\pi \tilde{P} = \pi \iff \pi Q = 0.$$

Thus π is also the unique probability solution of $\pi Q = 0$.

By the discrete-time convergence theorem, $\tilde{P}^n(x, y) \rightarrow \pi(y)$. Uniformization at rate a yields

$$H_t(x, y) = \sum_{n=0}^{\infty} e^{-at} \frac{(at)^n}{n!} \tilde{P}^n(x, y) = \mathbb{E}[\tilde{P}^{N_t}(x, y)],$$

where $N_t \sim \text{Poisson}(at)$. Fix $\varepsilon > 0$ and choose m so that $|\tilde{P}^n(x, y) - \pi(y)| < \varepsilon$ for every $n \geq m$. Since the absolute difference is at most 1 for $n < m$,

$$|H_t(x, y) - \pi(y)| \leq \mathbb{P}(N_t < m) + \varepsilon.$$

For fixed m ,

$$\mathbb{P}(N_t < m) = e^{-at} \sum_{n=0}^{m-1} \frac{(at)^n}{n!} \rightarrow 0,$$

because an exponential dominates every fixed polynomial. Hence $\limsup_{t \rightarrow \infty} |H_t(x, y) - \pi(y)| \leq \varepsilon$. Letting $\varepsilon \downarrow 0$ proves the convergence. $\square$

3 Discrete Time and Continuous Time at a Glance

	Discrete time	Continuous time
Clock	$n = 0, 1, 2, \dots$	$t \geq 0$
Local rule	transition matrix P	generator Q
Evolution	P^n	$H_t = e^{tQ}$
Stationarity	$\pi P = \pi$	$\pi Q = 0$
Composition	$P^{m+n} = P^m P^n$	$H_{s+t} = H_s H_t$
Uniformization	—	$Q = r(P - I)$

The central idea is that a continuous-time Markov chain consists of a jump chain together with random holding times. Uniformization places potential jumps at the times of a Poisson process and thereby connects continuous-time evolution to an ordinary discrete-time transition matrix. In discrete time, matrix powers describe evolution; in continuous time, the same role is played by the matrix exponential.